

24-12321 JAMES III T. AYALA

BCS224L-OCa Computer Organization & Assembly Language

2nd Sem SY 2025-26

8:00 AM - 11:00AM

Activity #15a GET/SET DATE

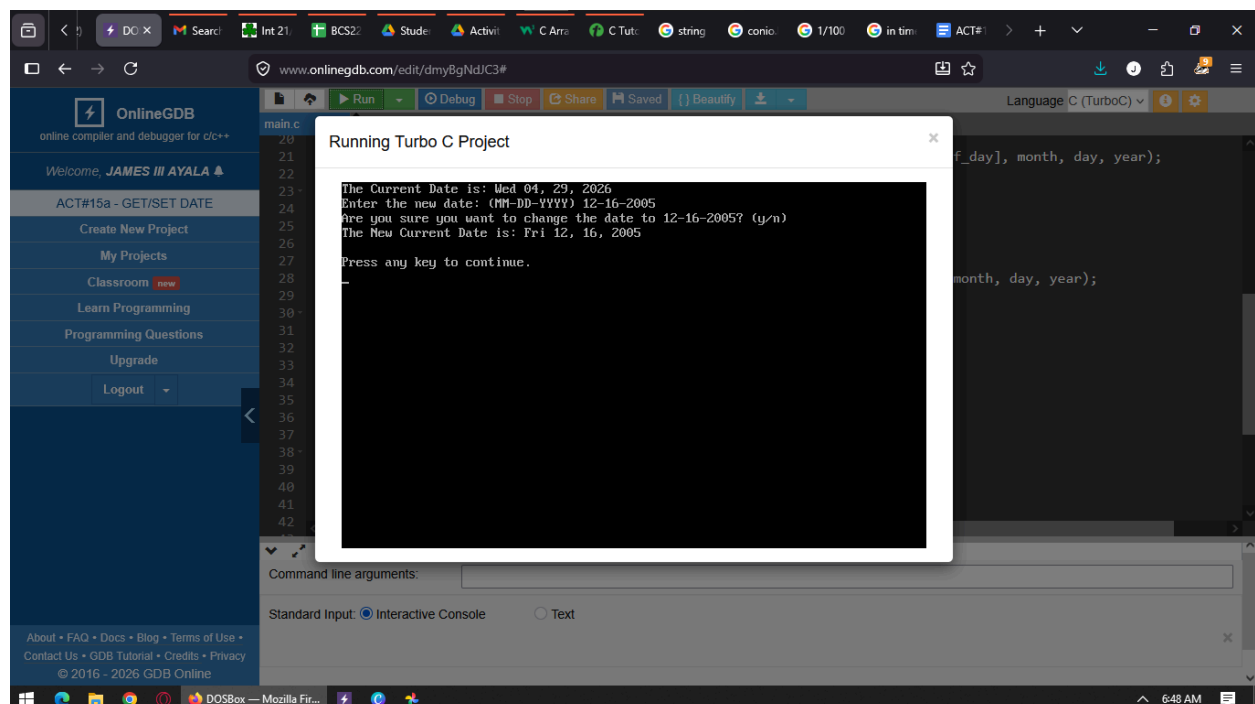
April 28, 2026

SOURCE CODE

```
#include <stdio.h>
#include <dos.h>
#include <conio.h>
void main(void){
    // Variables
    int month, day, year, bit_week_of_day;
    char confirm;
    // Convert week_of_day into a String
    const char *days_of_week[] = {"Sun", "Mon", "Tue",
    "Wed", "Thu", "Fri", "Sat"};
    union REGS inregs, outregs;
    // Access point of register GET DATE
    inregs.h.ah = 0x2A;
    // Access point of interrupt
    int86(0x21, &inregs, &outregs);
    clrscr();
    bit_week_of_day = outregs.h.al;
    month = outregs.h.dh;
    day = outregs.h.dl;
    year = outregs.x.cx;
    // Print the current date from the System
    printf("The Current Date is: %s %02d, %02d, %04d\n",
    days_of_week[bit_week_of_day], month, day, year);
    printf("Enter the new date: (MM-DD-YYYY) ");
    if(scanf("%d-%d-%d", &month, &day, &year) != 3) {
        printf("Error: Use the format MM-DD-YYYY (e.g.,
    12-25-2025)\n");
        printf("The system cannot accept the date entered");
        return;
    }
}
```

```
printf("Are you sure you want to change the date to
%02d-%02d-%04d? (y/n)", month, day, year);
confirm = getch();
if(confirm == 'y' || confirm == 'Y') {
    // Access point of register SET DATE
    inregs.h.ah = 0x2B;
    inregs.h.dh = month;
    inregs.h.dl = day;
    inregs.x.cx = year;
    // Access point of interrupt
    int86(0x21, &inregs, &outregs);
    if(outregs.h.al == 0) {
        // Access point of register GET Date
        // Calling it AGAIN TO UPDATE TO THE CHANGE
        DATE
        inregs.h.ah = 0x2A;
        int86(0x21, &inregs, &outregs);
        bit_week_of_day = outregs.h.al;
        printf("\nThe New Current Date is: %s %02d, %02d,
        %04d\n", days_of_week[bit_week_of_day], month, day,
        year);
    } else {
        printf("\nError: DOS rejected the date (Check for leap
        years or invalid ranges)\n");
    }
    } else {
        printf("\nOperation cancelled by user\n");
    }
}
```

OUTPUT:



The screenshot displays the OnlineGDB interface with a Turbo C project running. A modal window titled "Running Turbo C Project" shows the program's output. The output text is as follows:

```
The Current Date is: Wed 04, 29, 2026
Enter the new date: (MM-DD-YYYY) 12-16-2005
Are you sure you want to change the date to 12-16-2005? (y/n)
The New Current Date is: Fri 12, 16, 2005
Press any key to continue.
```

The background shows the OnlineGDB website with the user "JAMES III AYALA" logged in. The left sidebar contains navigation links like "Create New Project", "My Projects", "Classroom", "Learn Programming", "Programming Questions", "Upgrade", and "Logout". The bottom status bar indicates the command line arguments and standard input settings.